

Benchmarking gRPC and RESTful Services for Real-Time IoT System

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Runtimes for Concurrency and Distribution

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Presentation Outline

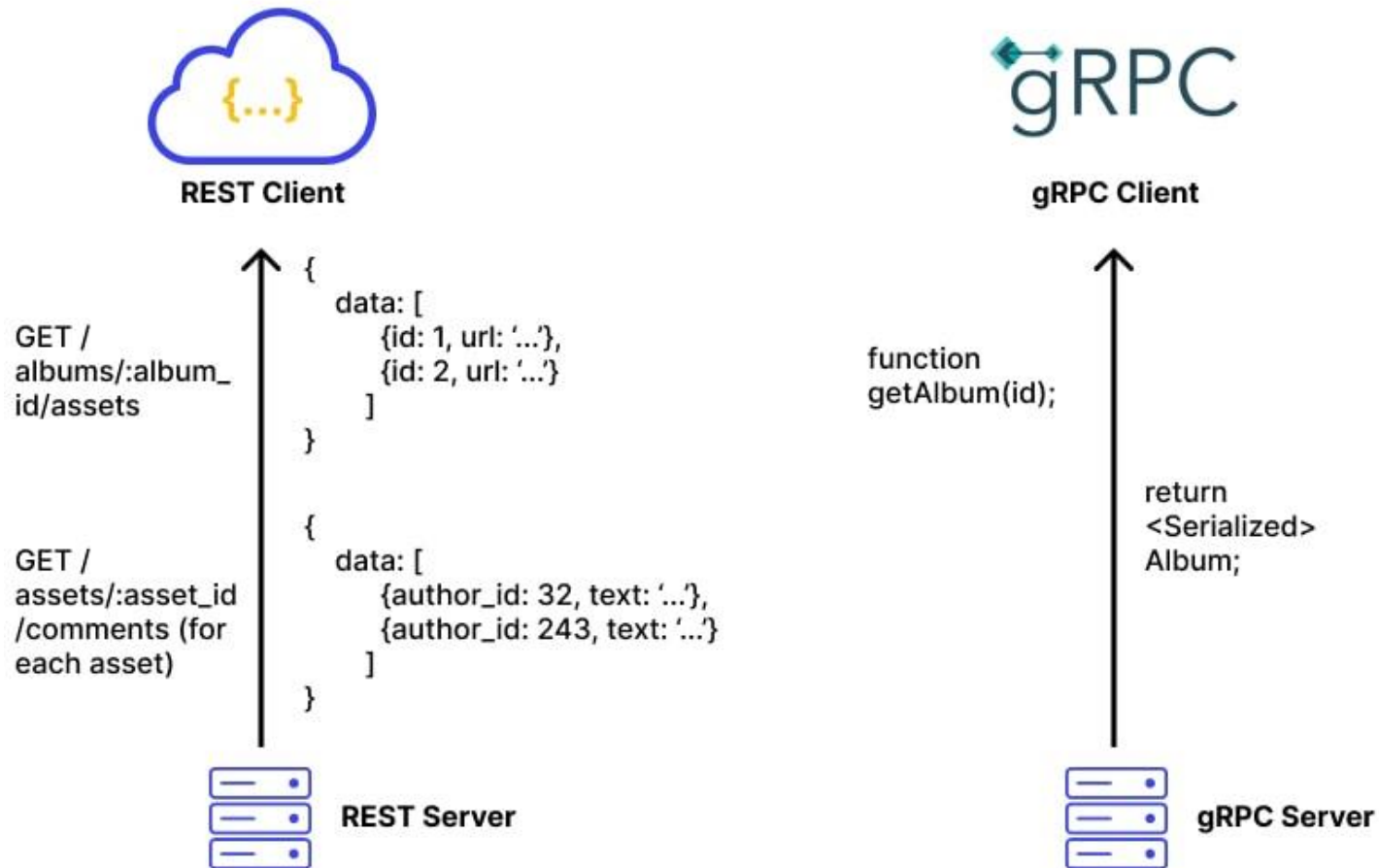


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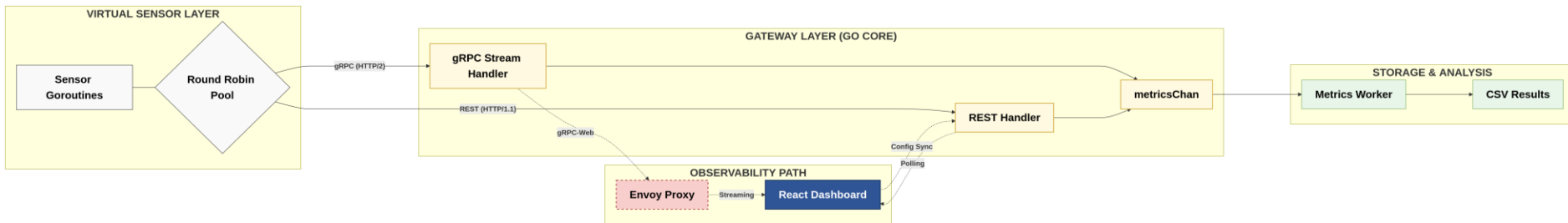
1. REST vs. gRPC
2. PoC Overview
3. Remarks and Corrections
4. Final Results
5. Learning Outcomes
6. Course Feedback



REST vs. gRPC



3-Layer Architecture Telemetry System



- **Virtual Sensor Layer:** for data generation
- **Gateway Layer:** dual-stack server for orchestration
- **Observability Layer:** web dashboard

Storage and analysis for benchmarking purposes

Remarks and Corrections

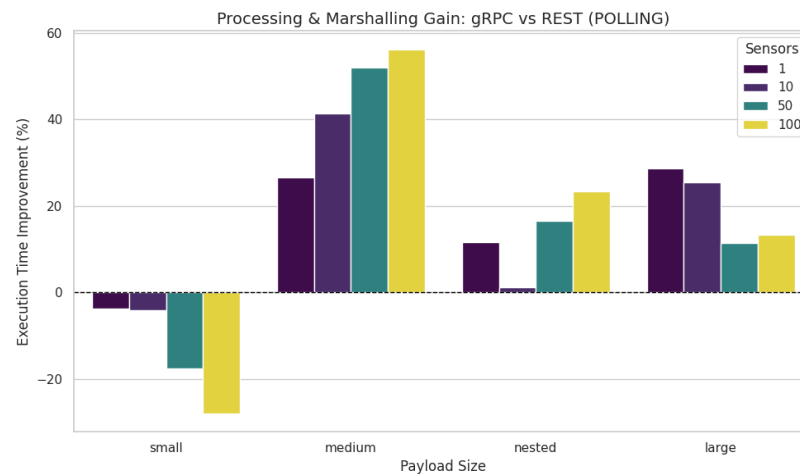
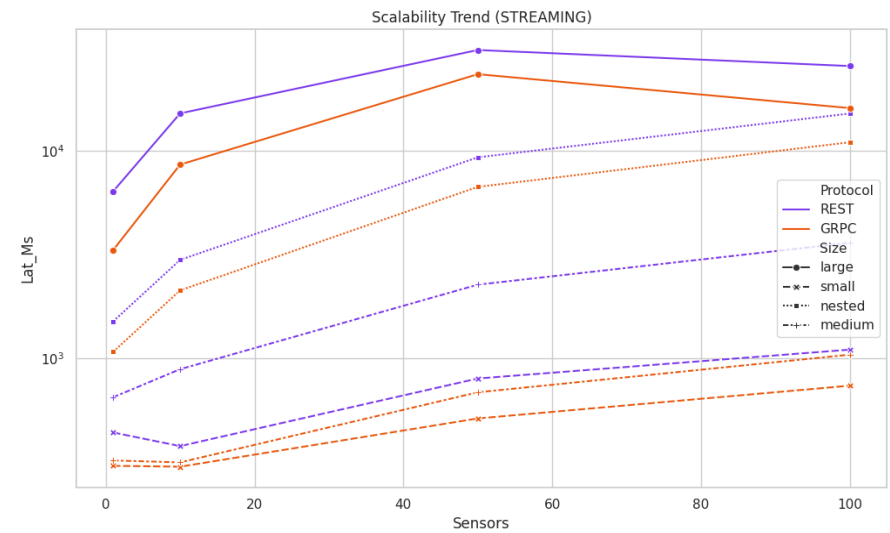
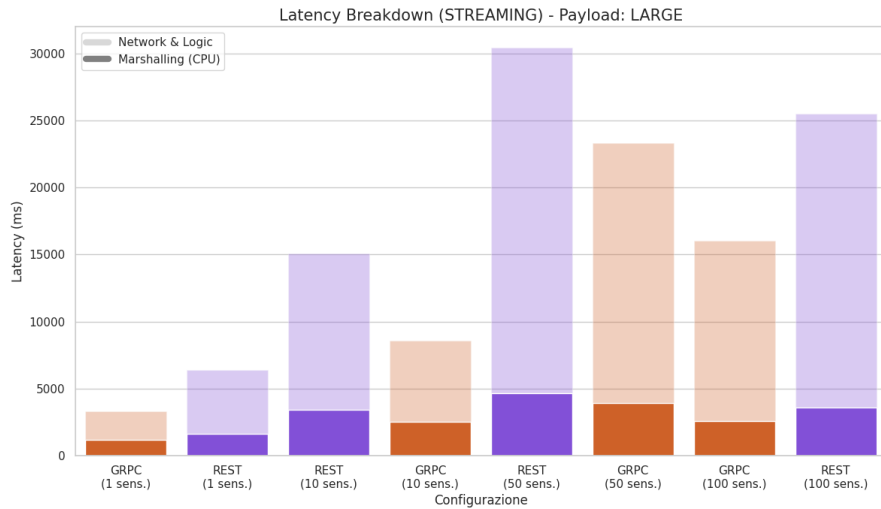
Confusing REST with HTTP/1.1

- REST is a resource-oriented architectural style, not a transport protocol;
- HTTP/1.1 and JSON are the most common implementation choice;
- REST vs. gRPC = Resources vs. Remote Procedures.

gRPC Browser Limitations

- Web browsers support HTTP/2 but block access to HTTP Trailing Headers;
- Trailing Headers are required by gRPC to transmit call status and error codes.

Final Results



- **Go Concurrency and Asynchronous Channels:** used Go runtime model for asynchronous Goroutines and created unbuffered and buffered asynchronous channels for I/O operations;
- **Distributed Networking Barriers:** found a solution to avoid Web APIs blocking access to Trailing Headers (Envoy Proxy)
- **Bundling and Runtime Distribution Barriers:** within the client application, troubleshooted Webpack configurations to handle incompatible modules that caused crashes at runtime;

- **Didactic Value and Engagement:** distinctive lecture style that challenges traditional habits, promoting critical conversation over passive learning;
- **Exam format:** hands-on approach with open project specifications;
- **Personal Retrospective:** started an ambitious PoC (maybe a little to ambitious) which caused time-management challenges.

Images Sources

- *Wallarm, "gRPC vs. REST: Comparing Key API Designs and Deciding Which One is Best."*

URL: <https://www.wallarm.com/what/grpc-vs-rest-comparing-key-api-designs-and-deciding-which-one-is-best>;

- Original architecture diagrams and benchmark charts were independently designed and generated via the local testing suite.